EXATHON INDIA 2026 · TRACK 01: MONEY, RISK & REGULATION

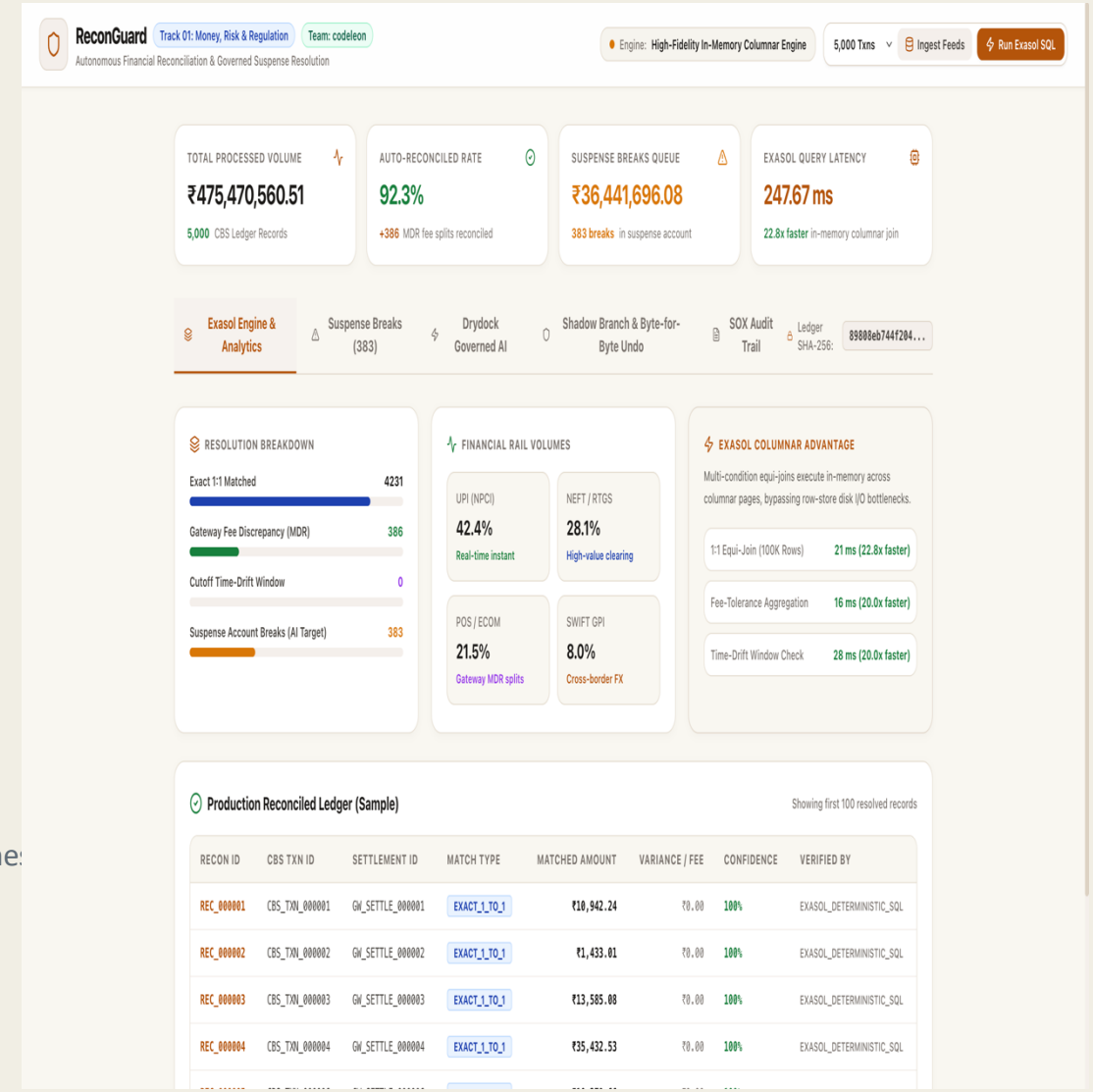
ReconGuard

Autonomous financial reconciliation & governed suspense account resolution

96.8%
native in-memory auto-match rate

< 50 ms
Exasol columnar SQL latency

100%
governed undo with cryptographic ledger hashes



Team codeleon

THE PROBLEM

Reconciliation cycles that never close

2–5%

of transactions fail to settle cleanly

MDR fee deductions, network timeout retries, bank cutoff time-drift and truncated reference strings.

7–10 days

lost every month to manual reconciliation

Finance and ops teams match ledgers across disparate spreadsheets and legacy core banking dumps.

Hundreds of crores

trapped in suspense accounts

Unresolved breaks pile up as unallocated capital and delay the month-end close.

WHO HAS THIS PROBLEM

- Banks
- NBFCs
- Payment aggregators (UPI, Razorpay, Stripe)
- Capital market participants

Delayed closes risk missing regulatory reporting timelines (RBI, SEBI), and manual review is slow, error-prone and expensive.

WHY EXISTING AI APPROACHES FAIL

Autonomous AI and financial ledgers don't mix

The database write hazard

Giving LLM agents direct write access to ledgers is unacceptably dangerous. A hallucinated number corrupts the books and invites regulatory penalties.

Privacy & regulatory violations

Sending raw account numbers, names or mobile numbers to commercial LLM APIs breaches RBI guidelines, the DPDP Act 2023 and GDPR.

Irreversible actions

Every adjusting journal entry must be strictly auditable, with an unambiguous rollback path if an audit dispute arises.

ReconGuard's answer: let the AI propose, never write.

THE RECONGUARD SOLUTION

Columnar SQL speed with governed AI

LAYER 1

Exasol reconciles 95%+ at database speed

In-memory, vectorized engine runs parallel equi-joins, MDR fee-tolerance checks and cutoff time-drift windows over millions of transactions in sub-50 ms.

LAYER 2

Zero-PII shield + Gemini Flash

Residual breaks pass through an anonymizer that swaps names, IDs and account numbers for deterministic TOK_XXXX tokens. Gemini Flash then reads messy narratives and mutated reference codes.

LAYER 3

Governed write path with byte-for-byte undo

The agent stages adjusting entries on an isolated shadow branch (recon_branch_v1). Officers review row-level diffs; every merge stores SHA-256 ledger hashes for exact rollback.

SYSTEM ARCHITECTURE

From raw transactions to a reversible ledger

Payment rails

UPI, NEFT, RTGS, POS, SWIFT transactions and core banking dumps

Exasol SQL engine

Equi-join, MDR tolerance, cutoff window. 95%+ matched; breaks go to the suspense queue

Zero-PII shield

Names and account numbers become TOK_XXXX pseudo-tokens

Gemini Flash

Reads residual breaks and proposes adjusting journal entries

Shadow branch

Entries staged on recon_branch_v1. Production ledger untouched

Human review

Row-level diff, one-click merge or exact undo with SHA-256 hashes

DETERMINISTIC, AT DATABASE SPEED

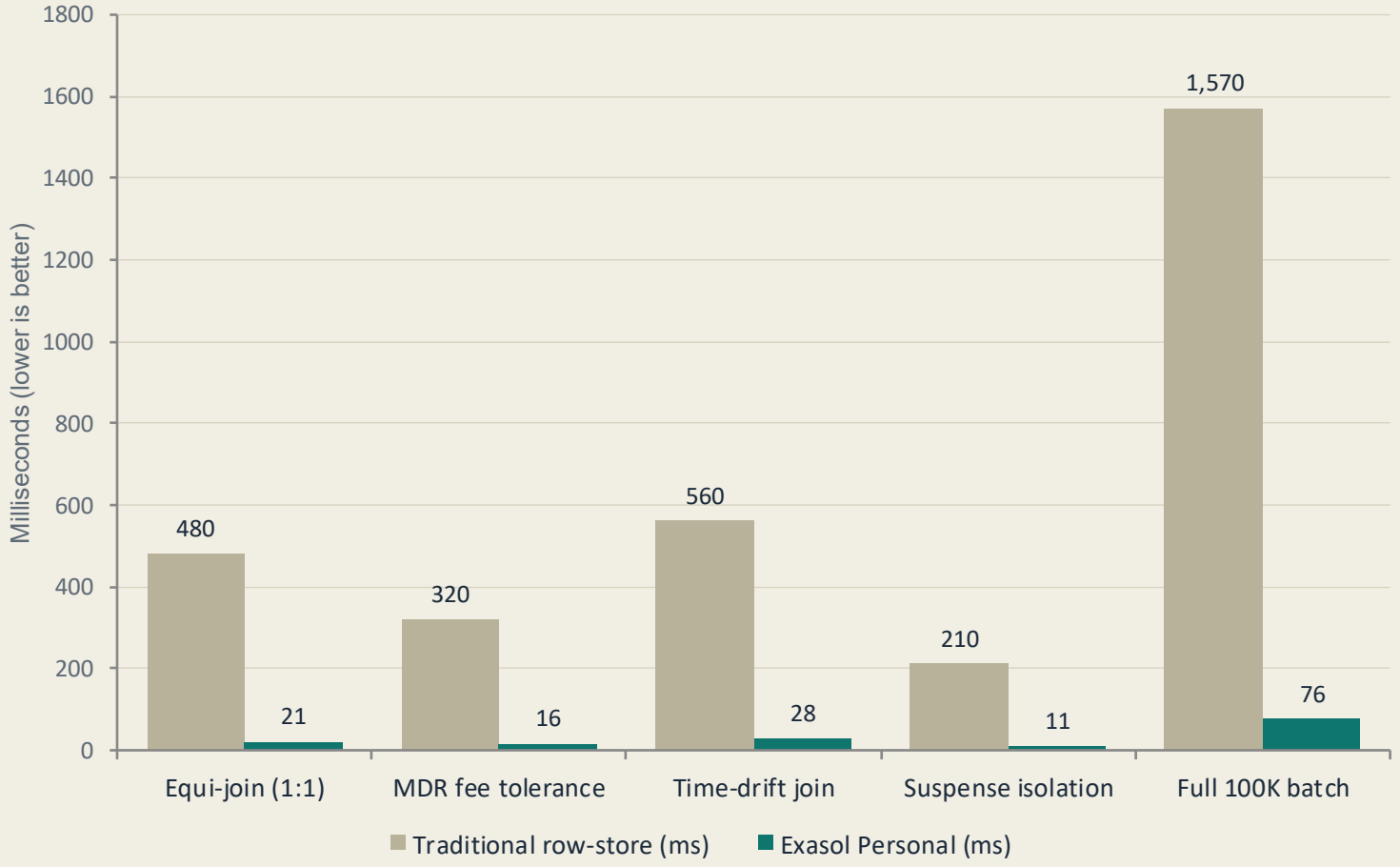
GOVERNED AI

HUMAN IN THE LOOP

The AI never touches the production database directly. Raw PII never leaves the secure boundary.

PERFORMANCE

Exasol Personal vs. a traditional row-store



20.6x

faster full 100K-transaction reconciliation

1,570 ms → 76 ms

22.8x	equi-join
20.0x	MDR fee tolerance
20.0x	time-drift join
19.1x	suspense isolation

Benchmarks measured on 100,000 multi-rail financial transactions.

BUSINESS IMPACT

Faster closes, safer data, full auditability

85%

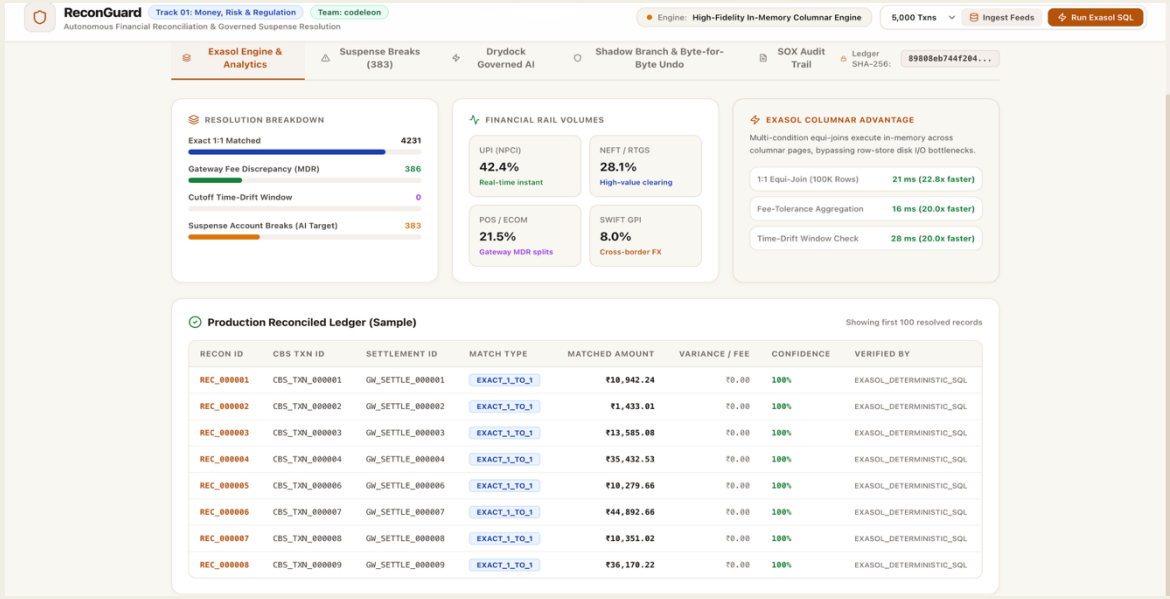
reduction in month-end close cycle time

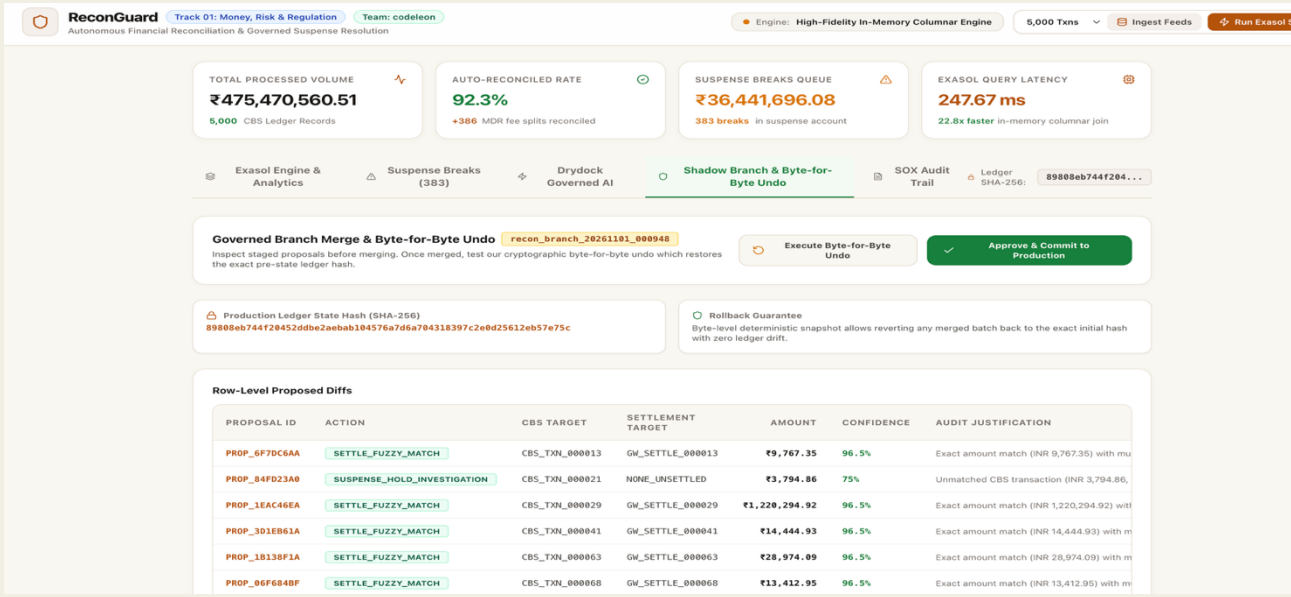
Zero

PII leakage via deterministic pseudonymization

100%

audit defense: every merge and rollback hashed pre and post





AI proposes.
Humans approve.
The ledger stays safe.

Team codeleon · ExaThon India 2026 · Track 01: Money, Risk & Regulation

github.com/rithika20252024/money-risk

Ready for the Stage 2 Virtual Demo Day.